UNIT - 1

Legal Framework and EIA Notification on Development and Environment

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Development activities such as industries, highways, dams, and mining can seriously affect air, water, land, ecosystems, and communities. To balance **economic growth** with **environmental protection**, governments have created a **legal framework** for Environmental Impact Assessment (EIA).
In **India**, this framework is mainly built around the **Environment (Protection) Act, 1986** and the **EIA Notification, 2006**, administered by the **Ministry of Environment, Forest and Climate Change**.

Legal Framework for Environment and Development

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1. Meaning of Legal Framework

The legal framework consists of:

- Acts and statutes
- Rules and notifications
- Regulatory authorities
- Judicial interpretations

It makes environmental assessment **mandatory** before approving development projects.

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2. International Influence on Environmental Law

Globally, EIA systems were encouraged by:

- **United Nations Environment Programme** — promoted environmental planning.
- **World Bank** — requires EIA for funded projects.

- **Organisation for Economic Co-operation and Development** — issued environmental guidelines.

These bodies shaped national laws on environment–development balance.

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3. Constitutional Support in India

The Constitution provides:

- Duty of the State to protect environment
- Duty of citizens to safeguard nature

Indian courts interpret the **right to life** as including the right to a clean environment.

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4. Environment (Protection) Act, 1986

This is the **umbrella legislation** for environmental protection in India. It empowers the Central Government to:

- Fix pollution standards
- Restrict industrial locations
- Regulate hazardous substances
- Order closure of polluting units
- Require environmental clearance

The EIA system legally flows from this Act.

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5. Supporting Environmental Laws

Development projects may also need approvals under:

- Forest conservation laws
- Wildlife protection laws
- Coastal regulation rules

These prevent damage to sensitive ecosystems.

EIA Notification, 2006 and Its Role in Development

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6. Meaning of EIA Notification

The **EIA Notification, 2006**, issued under the Environment (Protection) Act, makes **prior Environmental Clearance (EC)** compulsory for specified development projects.

It legally connects **environmental protection** with **development planning**.

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7. Categorization of Projects

Projects are classified into:

- **Category A** — large or high-impact projects; cleared at the national level.
- **Category B** — smaller projects; cleared at the state level.

This ensures that environmental scrutiny is proportional to project size and risk.

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8 Stages Prescribed in the Notification

The Notification defines a structured EIA process:

1. **Screening** — decides if detailed EIA is needed.
2. **Scoping** — fixes issues to be studied (Terms of Reference).
3. **Draft EIA Report** — impact analysis and mitigation plans.
4. **Public Consultation** — hearing affected people and stakeholders.
5. **Appraisal** — expert committee review and final decision.

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10. Authorities Under the Notification

Key implementing bodies include:

- Central expert committees under the national ministry
- State-level appraisal authorities
- Pollution Control Boards for hearings and monitoring

Importance for Development and Environment

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11. Controls Environmental Damage

Ensures pollution control, conservation of biodiversity, and protection of water resources.

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12. Promotes Sustainable Development

Balances infrastructure growth with long-term ecological safety.

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13. Ensures Accountability

Legal penalties, monitoring, and judicial review prevent irresponsible development.

Major Environmental Impacts of Development

Development activities such as industries, roads, dams, cities, mining, and power plants improve economic growth and living standards. However, when not properly planned and regulated, they can cause **serious environmental damage**.
The major impacts occur on **land, water, air, forests, wildlife, climate, and human health**.

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1. Deforestation and Loss of Vegetation

- Clearing forests for roads, housing, dams, and mining.
- Leads to soil erosion, landslides, and reduced rainfall.
- Destroys habitats of animals and birds.
- Reduces carbon absorption, contributing to climate change.

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2. Land Degradation and Soil Erosion

- Construction and mining remove topsoil.
- Over-irrigation and chemical use reduce soil fertility.

- Quarrying leaves scars on land.
- Desertification may occur in dry regions.

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3. Water Pollution

- Industrial effluents discharged into rivers and lakes.
- Sewage from growing cities contaminates water bodies.
- Agricultural runoff adds fertilizers and pesticides.
- Affects drinking water and aquatic life.

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4. Air Pollution

- Factories, thermal power plants, and vehicles release smoke and gases.
- Causes smog, acid rain, and respiratory diseases.
- Increases greenhouse gases like carbon dioxide.
- Reduces visibility and damages crops.

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5. Noise Pollution

- Traffic, airports, construction machines, and industries create high noise.
- Disturbs humans and wildlife.
- Causes stress, sleep disorders, and hearing problems.

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6. Loss of Biodiversity

- Destruction of forests and wetlands reduces species numbers.
- Fragmentation of habitats blocks animal movement.
- Some species become endangered or extinct.
- Ecosystem balance is disturbed.

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7. Climate Change and Global Warming

- Burning fossil fuels in industries and transport releases greenhouse gases.

- Deforestation reduces carbon sinks.
- Leads to rising temperatures, irregular rainfall, droughts, and floods.
- Melting glaciers and sea-level rise threaten coastal areas.

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8. Depletion of Natural Resources

- Excessive use of groundwater.
- Over-extraction of minerals and fossil fuels.
- Overfishing in seas and rivers.
- Makes resources scarce for future generations.

Historical Development of EIA

Environmental Impact Assessment (EIA) is a formal process used to **predict and evaluate environmental effects of development projects before approval**.
Its purpose is to prevent environmental damage, protect natural resources, and promote **sustainable development**.

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1. Early Environmental Awareness (Before 1970)

Rapid industrial growth and urbanization after World War II caused:

- Air and water pollution
- Deforestation
- Health problems
- Industrial accidents

Governments began realizing that **environmental protection must be part of development planning**, rather than reacting after damage occurred.

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2. Formal Birth of EIA (1970s)

The first systematic EIA system emerged in the United States in 1970 through federal environmental legislation.
This made environmental studies **mandatory for major public projects** and created a model for other countries.

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3. International Promotion of EIA

Soon, international agencies supported EIA adoption worldwide:

- **United Nations Environment Programme** promoted environmental planning in development.
- **World Bank** required EIA for projects it financed.
- **Organisation for Economic Co-operation and Development** issued environmental-assessment guidelines.

This phase spread EIA across developed and developing nations.

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4. Expansion in Developing Countries (1980s–1990s)

Many Asian, African, and Latin American countries adopted EIA because:

- Large infrastructure projects increased
- International funding agencies demanded safeguards
- Environmental awareness grew

Governments created:

- Environmental ministries
- Clearance systems
- Legal EIA procedures

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5. Modern EIA and Sustainable Development (1990s–Present)

After global discussions on sustainability, EIA expanded to include:

- Climate-change impacts
- Biodiversity conservation
- Social and health aspects
- Risk analysis
- Public participation
- Cumulative-impact studies

In countries such as **India**, EIA is now legally required for many development projects.

Types of Environmental Impact Assessment

Different types of EIA are applied depending on **planning level, scale, and purpose**.

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6. Project-Level EIA

- Most common type.
- Evaluates individual projects such as dams, highways, power plants, factories, and mines.
- Studies pollution, land use, wildlife loss, and social impacts.

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7. Strategic Environmental Assessment (SEA)

- Conducted for **policies, plans, and programmes**.
- Applied before specific projects are designed.
- Example: national transport or energy policy.

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8. Regional Environmental Assessment

- Focuses on a whole geographic area.
- Used when many projects exist in one region (e.g., industrial corridors or mining belts).
- Examines combined environmental stress.

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9. Sectoral EIA

- Studies impacts of an entire **industrial sector**, such as power, mining, transport, or irrigation.
- Helps frame standards and siting guidelines.

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10. Cumulative Impact Assessment

- Assesses total effects of **past, present, and future activities together**.
- Important when several projects affect the same river, forest, or air-shed.

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11. Rapid or Preliminary EIA

- Quick assessment during early planning.
- Used to decide whether a **full EIA** is required.

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12. Alternative-Based or Comparative EIA

- Compares:
  - Different locations
  - Technologies
  - Designs
  - “No-project” option
- Helps select the least-damaging option.

UNIT – 2

Checklist for ESIA

An ESIA checklist is a systematic tool used in the Environmental and Social Impact Assessment process to ensure that all important environmental and social aspects of a proposed project are identified and evaluated. It helps consultants and decision makers to avoid missing any key impact during assessment.

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1. Purpose of the ESIA Checklist

- **To provide a structured list of issues that need to be considered in ESIA**
- **To help identify potential environmental and social impacts**
- **To support impact prediction, evaluation, and mitigation planning**
- **To ensure consistency and completeness in assessment**
- **To act as a preliminary screening tool before detailed studies begin**

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2. Basic Components of an ESIA Checklist

A standard ESIA checklist typically includes the following sections:

A. Project Information

- **Project name, type, location**
- **Size, layout and operational stages**
- **Technology and raw materials used**

B. Environment and Ecosystem Factors

- **Air quality**
- **Water resources (surface and groundwater)**
- **Soil texture and erosion**
- **Noise and vibration**
- **Flora and fauna (biodiversity)**
- **Sensitive ecological zones**

C. Social and Human Factors

- **Population and settlement pattern**
- **Land use and land ownership**
- **Livelihood and employment**
- **Health and sanitation**
- **Cultural and heritage sites**
- **Community institutions**

D. Economic and Infrastructure Aspects

- **Existing infrastructure**
- **Local economic activities**
- **Transport and traffic**
- **Public utilities**

E. Legal and Compliance Issues

- **Applicable environmental laws**
- **Permits and legal clearances**
- **Standards for emissions and effluents**

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3. How a Checklist Helps in ESIA

A checklist aids the ESIA process by:

- **Prompting early identification of impacts**
- **Guiding data collection and baseline analysis**
- **Highlighting issues that need detailed study**
- **Supporting comparison of "before" and "after" impacts**
- **Serving as a reference for public consultation and reporting**

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4. Types of Checklists Used in ESIA

There are different types of checklists depending on how detailed they are:

1. **Simple Checklists**
    - **Yes/No questions**
    - **Quick screening**
    - **Good for small projects**
2. **Descriptive Checklists**
    - **More detailed explanations**
    - **Describes the *nature* of potential effects**
3. **Scaling or Scoring Checklists**
    - **Assigns numbers to impacts**
    - **Helps in ranking severity of impacts**
4. **Weighted Checklists**
    - **Impacts are weighted based on importance**
    - **Useful for prioritizing mitigation actions**

(These types are discussed in EIA methodology videos under checklist tools, though not directly quoted from your video)

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5. Example of a Simple ESIA Checklist Item

Check Item	Yes / No	Remarks
Does the project affect water bodies?	Yes/No	Identify type and distance
Will waste be generated?	Yes/No	Type and disposal method
Is there displacement of people?	Yes/No	Quantity and relocation plan
Are there protected forests nearby?	Yes/No	Forest clearance required

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Mathematical Models in Impact Prediction

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1. Meaning of Mathematical Model in ESIA

A mathematical model is a quantitative tool used to predict environmental impacts of a project by simulating interactions between environmental components. It uses equations, data, and algorithms to estimate how a project will affect air, water, soil, noise, etc.

Mathematical models are important in ESIA because they provide scientific, measurable, and reliable predictions of impacts before they occur.

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2. Role of Impact Prediction in ESIA

Impact prediction is a key step in ESIA and helps to:

- **Forecast future environmental changes**
- **Estimate severity of impacts**
- **Compare “with project” and “without project” scenarios**
- **Support decision making**
- **Plan effective mitigation measures”**

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3. What Mathematical Models Do

Mathematical models:

- **Use baseline data (existing environmental conditions)**
- **Apply mathematical equations**
- **Consider project parameters**
- **Output predicted values for environmental components**

For example, a model may calculate how much air pollution will increase due to a factory or how noise levels will rise near a highway.

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4. Examples of Models Used in ESIA

Some common types of models include:

1. **Air Quality Models**

Predict how emissions disperse in the atmosphere.

2. **Water Quality Models**

Estimate changes in water pollutant concentrations.

3. **Noise Prediction Models**

Calculate sound levels at different distances from sources.

4. **Hydrological Models**

Forecast changes in surface and groundwater flow.

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5. Steps in Using a Mathematical Model

1. **Data Collection**
    - **Baseline environmental data**
2. **Model Selection**
    - **Choose based on impact type (air, water, noise)**
3. **Input Parameters**
    - **Project specifications (e.g., emissions, fuel used)**
4. **Calibration**
    - **Adjust model with known data**

5. **Run Prediction**
    - **Calculate future conditions**
6. **Analysis of Results**
    - **Interpret predicted impacts**

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6. Advantages of Mathematical Models

- **Scientific and accurate forecasting**
- **Helps in quantification of impacts**
- **Improves credibility of ESIA**
- **Supports decision-making and planning**

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7. Limitations of Mathematical Models

- **Requires reliable data**
- **High technical expertise needed**
- **May need powerful computer tools**
- **Can be expensive**

UNIT - 3

Environmental Monitoring Plan for Various Natural Factors

An Environmental Monitoring Plan (EMP/Monitoring Plan) is a systematic program to measure environmental conditions before, during, and after a project.
It checks whether predicted impacts in the Environmental Impact Assessment (EIA) are correct and whether mitigation measures are working.

Objectives of Environmental Monitoring

- **Track changes in natural resources**
- **Ensure compliance with standards**
- **Detect pollution early**
- **Protect ecosystems and communities**
- **Support corrective actions**
- **Improve project management**

Monitoring Plan for Major Natural Factors

1. Air Environment

Parameters Monitored

- **Particulate matter (PM_{10}, $PM_{2.5}$)**
- **Sulphur dioxide (SO_2)**
- **Nitrogen oxides (NO_x)**
- **Carbon monoxide (CO)**
- **Volatile organic compounds (VOCs)**

Methods

- **Continuous air monitoring stations**
- **High-volume samplers**
- **Meteorological data collection (wind, temperature)**

Frequency

- **Weekly to monthly (construction phase)**
- **Quarterly or continuous (operation phase)**

Purpose

- **Check industrial and traffic emissions**
- **Assess health risk to nearby population**

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2. Water Environment

Surface Water & Groundwater Parameters

- **pH and temperature**
- **Dissolved oxygen (DO)**
- **Biochemical oxygen demand (BOD)**
- **Chemical oxygen demand (COD)**
- **Turbidity**
- **Heavy metals**
- **Oil and grease**

Methods

- **Grab sampling from rivers, lakes, wells**
- **Laboratory analysis**
- **Flow measurements**

Frequency

- **Monthly or seasonal**

Purpose

- **Detect contamination**
- **Protect drinking-water sources and aquatic life**

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3. Soil and Land Environment

Parameters

- **Soil pH and salinity**
- **Organic matter**
- **Nutrient levels**
- **Heavy metals**
- **Oil contamination**

Methods

- **Core sampling**
- **Laboratory testing**
- **Land-use surveys**

Frequency

- **Half-yearly or yearly**

Purpose

- **Identify land degradation**
- **Assess agricultural productivity**
- **Check contamination from waste disposal**

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4. Noise and Vibration

Parameters

- **Day and night sound levels (dB(A))**
- **Peak construction noise**
- **Ground vibration near blasting zones**

Methods

- **Sound-level meters**
- **Vibration monitors**

Frequency

- **Weekly during construction**
- **Quarterly during operation**

Purpose

- **Protect human health**
- **Reduce disturbance to wildlife**

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5. Biological Environment (Flora & Fauna)

Parameters

- **Species diversity**
- **Population counts**
- **Habitat condition**
- **Presence of endangered species**
- **Vegetation cover**

Methods

- **Field surveys and transects**
- **Camera traps**
- **Satellite imagery**
- **GIS mapping**

Frequency

- **Seasonal or annual**

Purpose

- **Track biodiversity loss or recovery**
- **Evaluate success of afforestation programs**